Behavior-Guided Multimodal Policy Search: Leveraging Vision-Language Models for Reinforcement Learning

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Abstract

Large language models (LLMs) have emerged as promising black-box optimizers for reinforcement learning (RL) policy search, yet existing approaches operate in a purely textual feedback loop, relying on scalar rewards, rankings, or natural-language summaries that reveal whether a policy succeeds or fails but provide little insight into why it fails. To address this limitation, we introduce Behavior-guided Multimodal Policy Search (BMPS), a framework that augments LLM-driven optimization with structured visual diagnostics. First, BMPS constructs a local behavioral landscape by perturbing policy parameters, extracting representative trajectory frames, and using vision-language models (VLMs) to identify failure modes, spatial patterns, and behavioral deviations. Then, we integrate this multimodal feedback with reward histories into the LLM prompt to guide policy search. Across discrete and continuous control tasks, BMPS consistently outperforms policy-gradient baselines, prior LLM-driven optimization methods, and evolutionary approaches, with the largest gains in environments with sparse or weak informative rewards. Taken together, these results show that grounding policy search in interpretable behavioral feedback makes LLM-based optimization more effective and diagnosis-aware.

Keywords

Reinforcement Learning, Policy Search, Large Language Models, Vision-Language Models, Multimodal Feedback, Black-Box Optimization

1 Introduction

Reinforcement Learning (RL) [31] addresses a fundamental problem in machine learning: learning decision-making policies through interaction with an environment. RL has achieved remarkable results across complex domains, from Atari games [18, 19] to Go, Chess, and Shogi [29, 30], and Dota 2 [24]. Traditional methods such as PPO [28] and TRPO [27] rely on scalar reward signals, which offer only sparse, evaluative feedback with no semantic structure, limiting interpretability and exploration efficiency. The rise of large language models (LLMs) has led to their increasing integration into RL pipelines, where they are primarily used as supporting components.

LLMs have recently been explored in RL both as auxiliary modules, such as reward shapers or sources of prior knowledge [35, 37], and as more general black-box optimizers [4, 36]. Nevertheless, when applied to RL policy search, existing LLM-based optimizers still rely almost exclusively on scalar reward signals. This form of feedback indicates whether a policy performs well or poorly, but it does not explain which concrete behaviors caused success or failure. As a result, the model receives only weak supervision for policy refinement: it may observe that one candidate is better than another, yet still lack the behavioral evidence needed to infer how the policy should change. This limitation is especially severe in environments with sparse, delayed, or deceptive rewards, where return values alone may obscure the underlying reasons for poor performance. Consequently, the LLM cannot fully leverage its strengths in semantic reasoning, comparison, and diagnosis, and policy search then becomes more like trial and error based on rewards, rather than a process of learning from clear behavioral insights.

To address these limitations, we propose Behavior-guided Multimodal Policy Search (BMPS), a novel framework for reinforcement learning policy optimization with multimodal diagnostic feedback. BMPS advances LLM-based policy search in two key ways. First, BMPS treats the LLM as a direct optimizer over policy parameters, allowing policy updates to be generated through iterative language-based reasoning rather than relying solely on conventional gradient-based optimization. Second, and most importantly, BMPS introduces visual feedback-driven behavioral analysis into the optimization loop. By leveraging vision-language models (VLMs) to examine representative trajectory frames beyond scalar rewards alone, BMPS enables targeted and diagnosis-aware policy refinement. This transforms LLM-based policy search from a largely reward-driven trial-and-error process into a more grounded and interpretable optimization procedure. In our evaluation, BMPS has higher mean returns than prior LLM-based optimizers on 12 of 13 tasks and ties on Maze, while comparisons with policy-gradient and evolutionary baselines vary by task. In particular, BMPS falls below the leading baselines on Hopper, Navigation, and Inverted Double Pendulum.

In summary, our contributions are as follows:

- **Behavior-guided multimodal policy optimization.** We propose a framework that augments LLM-based policy search with structured visual feedback through a novel neighborhood behavioral sampling mechanism. Specifically, BMPS perturbs policy parameters via Poisson sampling around anchor policies (current, best, worst), performs rollouts, and leverages vision-language models (VLMs) to analyze trajectory frames in parallel. This process constructs a local behavioral landscape that captures how small parameter variations affect observed behaviors, enabling diagnosis-aware optimization beyond scalar reward signals.
- **Regret analysis of behavior-guided search.** We show $O(\sqrt{T})$ Bayesian regret when the LLM approximates posterior sampling and bound the simple regret of the returned policy under a weak condition on the proposals. Both results are tied to behavioral feedback through a measurable feedback effect, whose expectation splits into the decision value of the descriptions and the proposal regret of the LLM.
- **Empirical evaluation across RL benchmarks.** We evaluate BMPS on 13 discrete- and continuous-control tasks,

comparing against policy-gradient baselines, prior LLM-driven optimization methods, and evolutionary approaches. BMPS achieves the best average rank and higher mean returns than the prior LLM-based methods on 12 tasks, tying on Maze.

2 Related Work

Classical reinforcement learning optimizes policies from scalar returns using gradient-based methods such as TRPO and PPO [27, 28] or derivative-free search [17], achieving strong results across sequential decision-making problems [31]. However, scalar feedback provides limited information about underlying behaviors and failure modes.

To enrich supervision, prior work incorporates language into RL. Early approaches used linguistic signals for exploration, transfer, and interpretability [6, 10, 22], but were limited in expressiveness and scalability [15]. With modern LLMs, free-form text enables broader integration into RL pipelines [35, 39]. Most existing methods, however, use language as auxiliary guidance rather than as the optimization mechanism itself. A prominent direction is reward synthesis, where LLMs translate human intent into reward functions or evaluative signals [11, 13, 16, 37]. For example, DrEureka designs rewards and environment variations but relies on PPO for optimization [16], while language model predictive control converts feedback into executable control programs [13]. These approaches require collapsing semantic feedback into scalar objectives, which can introduce ambiguity and reward misspecification [9].Other work integrates language more directly into decision-making, such as guiding actions [7], modifying value updates [34], or coordinating modules [14], but still does not treat LLMs as optimizers over policy parameters.

Our work instead studies LLMs as policy optimizers, closely related to in-context RL and algorithm distillation [12, 40], contextual bandit formulations [21], and the view of LLMs as black-box optimizers [4, 36], as well as recent policy refinement approaches [1, 5, 38]. In particular, Prompted Policy Search (ProPS) [38] treats the LLM as a direct policy optimizer that proposes policy parameters from parameter–reward histories, while ProPS+ further incorporates semantic task information such as environment descriptions, goals, and expert hints. These methods are especially relevant to our setting and directly inspire BMPS. However, both still rely primarily on textual and scalar feedback, without grounding optimization in trajectory-level behavioral analysis. BMPS extends this line of work by integrating VLM-based visual diagnostics with neighborhood policy sampling to construct a local behavioral landscape, enabling multimodal behavior-guided policy refinement beyond reward-only prompting.

3 Methodology

3.1 Problem Formulation

We consider an episodic environment with state space $\mathcal{S}$, action space $\mathcal{A}$, reward function r, and horizon H. A policy π_θ with parameters $\theta \in \Theta$ induces a distribution p_θ over trajectories τ, and the goal is to maximize the expected return

$$J(\theta) = \mathbb{E}_{\tau\sim p_\theta}\big[R(\tau)\big], \qquad R(\tau) = \sum_{h=0}^{H-1} r(s_h, a_h), \tag{1}$$

where an episode may terminate before the horizon. The optimizer treats J as a black box: it observes neither gradients nor a model of the dynamics. A policy proposed by the optimizer is evaluated by the Monte Carlo estimate

$$\widehat{J}(\theta) = \frac{1}{L}\sum_{i=1}^{L} R(\tau_i), \qquad \tau_1, \ldots, \tau_L \overset{\text{i.i.d.}}{\sim} p_\theta, \tag{2}$$

with $L = 20$ rollouts in our experiments.

Policy classes. For continuous state spaces, we use linear policies $\pi_\theta(s) = W^\top s + b$, where $\theta \in \mathbb{R}^d$ stacks the entries of W and b; for discrete action sets, the action with the largest linear score is executed. Each parameter is restricted to $[-6, 6]$ with one decimal place, which gives the grid $\Theta = \{-6, -5.9, \ldots, 6\}^d$ with 121^d elements. Despite their simplicity, linear policies are competitive when combined with random-search optimization [17]. For discrete state spaces, we use deterministic tabular policies $\pi_\theta(s) = \theta_s$ with $\theta_s \in \mathcal{A}$, so that $\Theta = \mathcal{A}^{|\mathcal{S}|}$ has $|\mathcal{A}|^{|\mathcal{S}|}$ elements. Both classes are finite; the regret bounds in Section 4 use finiteness only through $\log|\Theta|$ and do not rely on the grid structure.

Prompted policy search. At iteration t, the optimizer holds the history $\mathcal{H}_{t-1} = \{(\theta_k, \widehat{J}(\theta_k))\}$ of all evaluated policies, including W randomly sampled warm-up policies, and a textual task description c. A reward-only prompted optimizer such as ProPS [38] draws the next policy from $P_{\text{LLM}}(\cdot \mid \mathcal{H}_{t-1}, c)$, the distribution that the LLM and its decoding procedure induce on policies once the response is parsed into a parameter vector. We call any such prompt-induced distribution a *proposal distribution*. Except in Theorem 4.4, which assumes that it approximates posterior sampling, we make no assumption about its form; in particular, the LLM is not assumed to perform Bayesian inference. BMPS enriches the prompt with a behavioral landscape, described next.

3.2 Behavior-guided Multimodal Policy Search

BMPS augments prompted policy search with a *behavioral landscape*: a set of rollouts around selected anchor policies, each paired with a VLM description of what the agent did. Each iteration consists of the steps below (Figure 1). Figure 2 illustrates how this evidence enters proposals in a logged search run.

At each iteration t, the algorithm maintains a history H_{t-1} of all previously evaluated policies and their associated rewards, along with two behavioral anchors: the best-performing policy $\theta_{\text{best}} = \arg\max_{\theta_k} r_k$ and the worst-performing policy $\theta_{\text{worst}} = \arg\min_{\theta_k} r_k$ observed so far. The current policy θ_{curr} is initialized to the final randomly sampled policy from the warm-up phase and subsequently updated to θ_t at the end of each iteration. The current anchor connects diagnostics to the latest proposal, the best retains an empirically successful reference, and the worst supplies examples of failure. Together, they provide contrasting behaviors within the same policy class, although their selection by estimated returns need not reflect the true ordering.

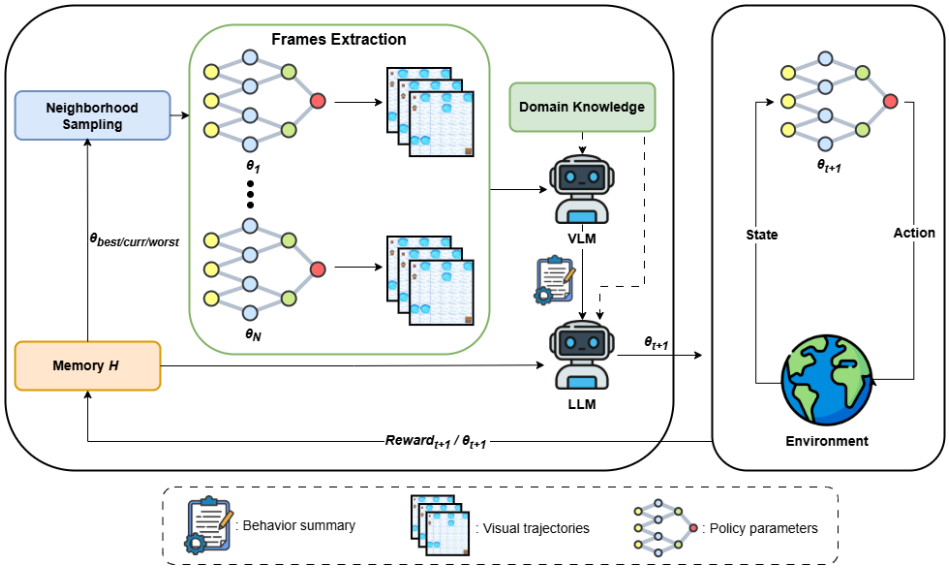


Figure 1: Overview of Behavior-guided Multimodal Policy Search (BMPS). At each iteration, the current policy is evaluated through environment rollouts, representative trajectory frames are extracted, and a vision-language model generates structured behavioral feedback. This multimodal feedback, together with reward statistics and policy history, is provided to the large language model to propose improved policy parameters for the next round of search.

Neighborhood Sampling. To build a local picture of the *behavioral landscape*, we sample N neighbor policies via Poisson perturbation around each of the three reference points $\theta \in \{\theta_{\text{best}}, \theta_{\text{worst}}, \theta_{\text{curr}}\}$. Given a reference parameter vector θ, each neighbor $\tilde{\theta}$ is generated by perturbing every dimension d by $\delta_d = s_d \cdot \sigma_d \cdot \epsilon$, where $s_d \sim \text{Poisson}(\lambda)$ is a non-negative integer step count, $\sigma_d \sim \text{Uniform}(\{-1, +1\})$ is a random sign, and ϵ is a fixed step size. The Poisson distribution is chosen deliberately: it concentrates mass near zero to favor small, local perturbations while its tail permits occasional larger jumps, naturally balancing local exploitation with broader exploration around each reference point. The integer step count fits the discrete parameter representation and allows some coordinates to remain unchanged. Since nearby parameters can still yield different behaviors, we use the rollouts as comparative evidence rather than interpret the perturbations as gradient estimates; useful exploration is not guaranteed.

Multimodal Analysis. The distinguishing feature of our approach is the use of a VLM to analyze the visual content of the collected trajectories, using either a joint VLM call or per-trajectory VLM summaries. For each trajectory, we sample one frame every 50 environment steps until the trajectory terminates. Given the sampled frames from the current policy rollout, the best and worst anchor trajectories, and their respective neighborhood rollouts, the VLM produces a structured behavioral summary ψ_t. This summary captures qualitative differences in agent behavior across trajectories-such as spatial positioning, motion patterns, object interactions, and failure signatures-that are difficult or impossible to encode through reward alone. By grounding the analysis in raw visual observations, the model can identify behaviorally meaningful distinctions that scalar metrics obscure. Separating visual diagnosis from policy proposal provides an inspectable text interface that the optimizer can combine with numerical evidence. Per-trajectory summaries also allow diagnostic calls to run in parallel. This compression can miss brief events or misinterpret observations, so it is a trade-off rather than an assumed advantage over direct image input.

Language-guided Policy Proposal. Conditioned on the history H_{t-1} and the multimodal analysis ψ_t, the LLM proposes a new candidate policy θ_t. This proposal step functions as an informed, semantics-aware mutation: rather than sampling blindly from a fixed perturbation kernel, the LLM uses its understanding of the behavioral summaries to suggest a policy that steers toward the characteristics of successful trajectories while explicitly departing from the patterns associated with failure. The history H_{t-1} provides additional context about which regions of the policy space have been explored and what reward signals they yielded, allowing the LLM to use both past successes and failures when generating the next proposal. Associating parameter changes with both descriptions and returns gives the LLM concrete alternatives to compare, while retaining history provides a record of earlier trials beyond the current neighborhood. These comparisons may inform a proposal, but a plausible behavioral explanation does not itself establish an improvement.

Evaluation and History Update. The proposed policy θ_t is then evaluated by collecting its estimated return $r_t = \hat{J}(\theta_t)$ from L fresh rollouts, and the history is extended to $H_t = H_{t-1} \cup \{(\theta_t, r_t)\}$. The anchors θ_{best} and θ_{worst} are updated accordingly if the new evaluation yields a more extreme reward in either direction. Single-rollout neighborhoods limit the cost of collecting diagnostic comparisons, whereas fresh evaluation rollouts reduce the influence of an isolated

favorable outcome on the recorded return. Neighborhood rollouts are not added to the selection history, and the returned policy is chosen from the evaluated proposals and warm-up policies, with finite-sample selection error still possible.

Together, these steps combine visual diagnostics, numerical returns, and search history. Behavioral similarity need not correspond to proximity in parameter space, so the landscape supplies comparisons of observed outcomes rather than a smooth local model of return. Section 4 distinguishes the information available in this feedback from the error incurred when the optimizer uses it.

4 Theoretical Analysis

Setup. We assume that returns lie in a known range $[r_{\min}, r_{\max}]$, let $B = r_{\max} - r_{\min}$, and use the normalized return $g(\tau) = (R(\tau) - r_{\min})/B \in [0,1]$ and its expectation $j(\theta) = \mathbb{E}_{\tau\sim p_\theta}[g(\tau)]$, so that $J = r_{\min} + Bj$. Let $C \subseteq \Theta$ be a finite set that contains every policy the parsing step can produce. At iteration t, the LLM receives the prompt $Z_t = (X_t, \Psi_t)$. The numerical context $X_t = (\mathcal{H}_{t-1}, \mathcal{N}_t, c)$ contains the history, the parameters and single-rollout returns $\mathcal{N}_t$ of all anchors and neighbors, and the task description, and Ψ_t contains the VLM descriptions of these rollouts. Since an LLM defines a conditional distribution over its outputs [2], BMPS draws its proposal from $Q_t = P_{\mathrm{LLM}}(\cdot \mid X_t, \Psi_t)$, whereas a numerical-only proposal at the same context would follow $P_t = P_{\mathrm{LLM}}(\cdot \mid X_t)$. Both distributions can be sampled, but their probabilities are never computed. Prompts are finite strings and take countably many values. We write $Z_{1:t} = (Z_1, \ldots, Z_t)$; the record Z_{t+1} contains θ_t and its estimated return, and Z_{T+1} is the record after the last evaluation.

Assumption 4.1 (Prompt-only proposals). *Conditionally on Z_t, the proposal θ_t is independent of the environment and of all other quantities generated before the LLM is queried at iteration t, and its distribution is Q_t.*

This holds by construction when the LLM sees only the prompt, its decoding randomness is drawn afresh, and the served model does not change during the run. The *cumulative regret* of the proposals is $\sum_{t=1}^{T}(j(\theta^*) - j(\theta_t))$, where θ^* is a best policy, and the *simple regret* is $j(\theta^*) - j(\widehat{\theta}_T)$ for the policy $\widehat{\theta}_T$ that the search returns.

4.1 Analytical Model and the Feedback Effect

Assumption 4.2 (Analytical task model). *The environment instance ξ is drawn from an arbitrary distribution, $j_\xi(\theta) \in [0,1]$ is its normalized expected return, and BMPS runs under ξ, so that ξ, the prompts, and the proposals have a joint distribution. We let θ^* be a C-valued random variable with $j_\xi(\theta^*) = \max_{\theta\in C} j_\xi(\theta)$, with ties broken by a fixed rule.*

This assumption is a modeling device, not a restriction on the system. The distribution of ξ need not be correctly specified, the LLM is assumed neither to know it nor to perform Bayesian inference, and the results below hold for every such distribution. The *Bayesian regret* of the proposals is

$$\mathrm{BR}_T = \mathbb{E}\Big[\sum_{t=1}^{T}\big(j_\xi(\theta^*) - j_\xi(\theta_t)\big)\Big]. \tag{3}$$

The feedback effect. Behavioral feedback enters the analysis through one measurable quantity. For an environment ξ and a context $z = (x, \psi)$, let $\mu_\xi^Q(z) = \mathbb{E}_{\theta\sim Q(\cdot|z)}[j_\xi(\theta)]$ and $\mu_\xi^P(z) = \mathbb{E}_{\theta\sim P(\cdot|x)}[j_\xi(\theta)]$, where $Q(\cdot \mid x, \psi) = P_{\mathrm{LLM}}(\cdot \mid x, \psi)$ and $P(\cdot \mid x) = P_{\mathrm{LLM}}(\cdot \mid x)$, and let $\mu_\xi^S(z)$ be the same quantity for a prompt in which the descriptions are randomly permuted across the rollouts of the landscape, with a fresh permutation for every query. The *feedback effect* and its pairing and text parts are

$$\Delta_\xi(z) = \mu_\xi^Q(z) - \mu_\xi^P(z) = \underbrace{\mu_\xi^Q(z) - \mu_\xi^S(z)}_{\Delta_\xi^{\mathrm{pair}}(z)} + \underbrace{\mu_\xi^S(z) - \mu_\xi^P(z)}_{\Delta_\xi^{\mathrm{text}}(z)}. \tag{4}$$

At a stored context of a BMPS run, $\Delta_\xi(z)$ is estimated without bias by querying the LLM n times with each prompt and evaluating each proposal with a fresh rollout, and by Hoeffding's inequality the estimate exceeds $\Delta_\xi(z)$ by more than $\sqrt{\log(1/\delta)/n}$ with probability at most δ. Proposition D.1 in Appendix D.1 relates the effect to the decision that the descriptions inform: $\mathbb{E}[\Delta_\xi(Z_t) \mid X_t = x] = \mathcal{I}_t(x) - \mathcal{R}_{1,t}(x) + \mathcal{R}_{0,t}(x)$. Here the *decision value* $\mathcal{I}_t \geq 0$ is what an ideal decision maker gains by also seeing the descriptions, and it is positive only if the descriptions can overturn the decision that the numerical evidence supports, while $\mathcal{R}_{1,t}$ and $\mathcal{R}_{0,t}$ are the proposal regrets of the LLM with and without them. A negative effect therefore reveals a loss in how the LLM uses the descriptions rather than a lack of information in them. Since Assumption 4.2 allows any distribution of ξ, taking it to be the distribution of the experimental tasks and seeds makes the average of such estimates over contexts of independent runs an unbiased estimate of the average gain in expected return of BMPS's proposals over numerical-only proposals made at the same contexts. The shuffled prompt has the length and the descriptions of the BMPS prompt but not their pairing with parameters and returns. Shuffled descriptions are a garbling of the paired ones [3], so their decision value lies between 0 and $\mathcal{I}_t$: $\Delta_\xi^{\mathrm{pair}}$ isolates what the anchor–neighbor pairing contributes, and $\Delta_\xi^{\mathrm{text}}$ what adding behavioral text contributes, including any harm from a longer prompt.

4.2 Sublinear Bayesian Regret

The feedback effect concerns a single iteration and does not by itself imply sublinear growth of BR_T. No such guarantee can hold without a condition on the LLM, since an LLM that always proposes the same suboptimal policy has linear regret. We first isolate the quantity that determines the growth rate. Let $H(\theta^* \mid A)$ be the entropy of the conditional distribution of θ^* given an event A, and let $H(\theta^* \mid A, Y)$ be its average over the values of a random variable Y (Appendix C). For a record $z_{1:t}$ of positive probability, let $A = \{Z_{1:t} = z_{1:t}\}$, and define the expected regret of iteration t, the information that it reveals about the best policy, and the information ratio [25],

$$\begin{aligned} \varrho_t(z_{1:t}) &= \mathbb{E}\big[j_\xi(\theta^*) - j_\xi(\theta_t) \,\big|\, A\big], \\ \iota_t(z_{1:t}) &= H(\theta^* \mid A) - H(\theta^* \mid A, Z_{t+1}), \\ \Gamma_t(z_{1:t}) &= \varrho_t(z_{1:t})^2/\iota_t(z_{1:t}), \end{aligned} \tag{5}$$

with $0/0 = 0$ and $a/0 = +\infty$ for $a > 0$. The quantity $\iota_t \geq 0$ is the conditional mutual information between θ^* and the next record, which contains the evaluation of θ_t and the next neighborhood with its descriptions.

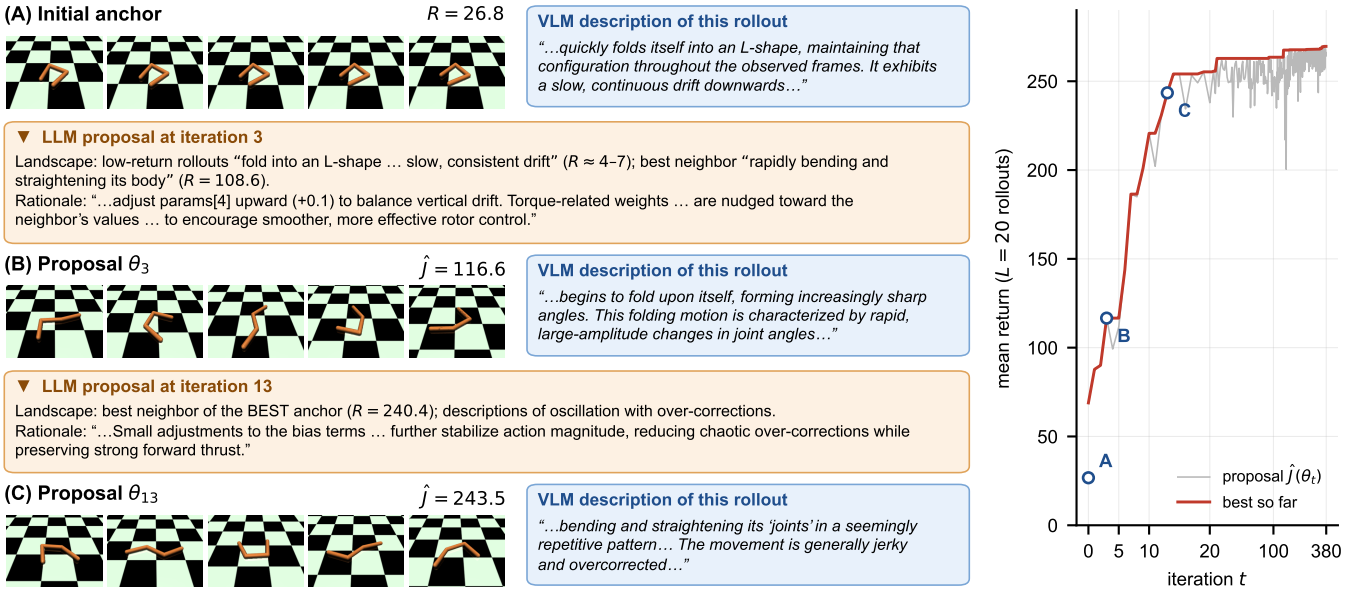


Figure 2: Behavior-guided search on *Swimmer* (one BMPS run). (A)–(C): rollouts of the initial anchor and of the proposals at iterations 3 and 13, each with a verbatim excerpt of the VLM description of that rollout. Orange boxes: excerpts of the behavioral landscape that the LLM received and of the rationale it returned with its proposal. R is the return of the described rollout, $\hat{J}$ the mean over $L = 20$ evaluation rollouts. Frames are re-rendered from the logged parameters at steps $200, 400, \ldots, 1000$. Right: evaluated return of each proposal and the best return so far (symmetric-log horizontal axis).

The analysis of [25] applies to any such loop. The information gains telescope, $\sum_t \mathbb{E}[\iota_t(Z_{1:t})] \le H(\theta^* \mid Z_1) \le \log|C|$, and the Cauchy–Schwarz inequality gives $\mathrm{BR}_T \le \sqrt{\overline{\Gamma}_T\, T \log|C|}$, where $\overline{\Gamma}_T$ is the average of $\mathbb{E}[\Gamma_t(Z_{1:t})]$ over $t \le T$. Search loops over C therefore differ only through the squared regret that they incur per unit of information about θ^*. Behavioral feedback acts on both terms of this ratio. For a fixed proposal at iteration t, recording Ψ_{t+1} in addition to X_{t+1} can only increase ι_t, because conditioning reduces entropy (Lemma C.2). The descriptions also change the proposal: if it were drawn from the numerical prompt, the regret of iteration t would change by $\mathbb{E}[j_\xi(\theta_t) - j_\xi(\theta_t^P) \mid Z_{1:t}]$, whose expectation is $\mathbb{E}[\Delta_\xi(Z_t)]$. Feedback lowers the ratio when both changes are favorable, and neither the theory nor a single measurement establishes both. The argument is also not specific to vision, since any additional record increases ι_t. A rate requires a condition on the LLM.

Let $\pi_t(\theta) = \Pr(\theta^* = \theta \mid A)$ be the posterior of the best policy given the record; Thompson sampling [33] would propose $\theta_t \sim \pi_t$. For a candidate θ with $\Pr(A, \theta_t = \theta) > 0$, let $A_\theta = A \cap \{\theta_t = \theta\}$, let $\iota_t(\theta; z_{1:t}) = H(\theta^* \mid A_\theta) - H(\theta^* \mid A_\theta, Z_{t+1})$ be the information that the next record reveals when θ is proposed, and let $\varrho_t^\pi(z_{1:t}) = \mathbb{E}[j_\xi(\theta^*) \mid A] - \sum_{\theta \in C} \pi_t(\theta)\, \mathbb{E}[j_\xi(\theta) \mid A]$ be the expected regret of Thompson sampling.

Assumption 4.3 (Approximate posterior sampling). *There are $c \ge 1$ and $\varepsilon_t(z_{1:t}) \in [0,1]$ such that, for every t and every record $z_{1:t}$ of positive probability, the proposal distribution $Q_t = P_{\mathrm{LLM}}(\cdot \mid z_t)$ satisfies $Q_t(\theta) \ge \pi_t(\theta)/c$ for every $\theta \in C$, and $\mathrm{TV}(Q_t, \pi_t) \le \varepsilon_t(z_{1:t})$.*

The first condition requires that the LLM not rule out policies that may be optimal given the record, and the second that it approaches posterior sampling as $\varepsilon_t \to 0$. Neither condition can be verified from samples alone, and Assumption 4.3 is used only in the following theorem.

Theorem 4.4 (Posterior sampling). *Let Assumptions 4.1–4.3 hold, and suppose that, for every record $z_{1:t}$ of positive probability,*

$$\varrho_t^\pi(z_{1:t})^2 \le \Gamma^\circ \sum_{\theta \in C} \pi_t(\theta)\, \iota_t(\theta; z_{1:t}). \tag{6}$$

Then

$$\mathrm{BR}_T \le \sqrt{c\,\Gamma^\circ\, T \log|C|} + \sum_{t=1}^{T} \mathbb{E}\big[\varepsilon_t(Z_{1:t})\big]. \tag{7}$$

If C is partitioned into M groups such that, whenever θ_t lies in a group G, the record Z_{t+1} contains, for each $\theta' \in G$, a statistic in $[0,1]$ whose conditional mean given ξ and the past is $j_\xi(\theta')$, then Eq. (6) holds with $\Gamma^\circ = M/2$.

The bound is sublinear whenever $\sum_t \mathbb{E}[\varepsilon_t] = o(T)$. The constant $\Gamma^\circ = M/2$ interpolates between bandit feedback, where only the return of the proposal is revealed ($M = |C|$, which recovers the ratio $|C|/2$ of Thompson sampling [25]), and full information ($M = 1$). Under the idealized group-feedback condition, observing unbiased statistics for a whole group of policies yields $M/2$ with M the number of groups, and this is how the behavioral landscape would enter the rate. For the grids of Section 3.1, $|C| = 121^d$, so the bound is informative only when each iteration reveals returns over large regions; the group condition idealizes anchor-centered sampling and is not a property that BMPS guarantees.

4.3 Quality of the Returned Policy

In policy search, intermediate proposals are evaluations rather than deployments, and what matters is the returned policy. BMPS returns the evaluated policy $\widehat{\theta}_T$ with the best recorded return, and for its simple regret a much weaker condition than Assumption 4.3 suffices. Fix the environment, let θ^* maximize j over Θ, let $\mathcal{S}_\varepsilon = \{\theta : j(\theta) \geq j(\theta^*) - \varepsilon\}$ be the set of ε-optimal policies, and let $q_t = Q_t(\mathcal{S}_\varepsilon)$ be the probability that the proposal of iteration t is ε-optimal.

Theorem 4.5 (Simple regret of the returned policy). *Let Assumption 4.1 hold, and suppose that each of the K evaluated policies, among them $\theta_1, \ldots, \theta_T$, has a recorded normalized return computed from L fresh rollouts. For every $s > 0$ and $\delta \in (0, 1)$, with probability at least $1 - e^{-s} - \delta$, either $\sum_{t=1}^{T} q_t < s$ or*

$$j(\theta^*) - j(\widehat{\theta}_T) \leq \varepsilon + 2\sqrt{\frac{\log(2K/\delta)}{2L}}. \tag{8}$$

Moreover, $q_t \geq 1 - r_t/\varepsilon$ for $\varepsilon > 0$, where $r_t = j(\theta^*) - \mathbb{E}_{Q_t}[j]$ is the expected regret of the proposal of iteration t.

The theorem tolerates arbitrarily many poor proposals. If the proposals keep a nonvanishing probability of being ε-optimal, say $q_t \geq q$, then $\sum_t q_t \geq qT$, and the returned policy is ε-optimal up to the estimation error with probability at least $1 - e^{-qT} - \delta$. Because K grows with T, the number L of evaluation rollouts must grow logarithmically to keep the estimation term fixed. Behavioral feedback enters through q_t: by the last statement, the lower bound on q_t increases as the expected proposal regret decreases. In the fixed environment of the theorem, this link to the measurement is direct and needs no distribution over environments. The expected proposal regrets with and without descriptions satisfy $r_t = r_t^P - \Delta_\xi(Z_t)$, where $r_t^P = j(\theta^*) - \mathbb{E}_{P_t}[j]$ and Δ_ξ is the feedback effect of Eq. (4), so that

$$q_t \geq 1 - \big(r_t^P - \Delta_\xi(Z_t)\big)/\varepsilon. \tag{9}$$

5 Experiments

5.1 Experimental Setting

Benchmark suite. We evaluate BMPS on a diverse suite of Gymnasium environments spanning both continuous and discrete control tasks. The benchmark is designed to provide broad coverage across different control settings, enabling a comprehensive evaluation of reasoning-driven policy search.

Compared baselines. We compare BMPS against a diverse set of baselines spanning policy gradient, evolutionary strategies, and prior LLM-based approaches. Our work is directly inspired by ProPS and ProPS+ [38], which pioneered the use of LLMs as black-box optimizers for RL policy search. ProPS represents the foundational variant, wherein the LLM is prompted to refine policies using only interaction history (state-action-reward sequences). ProPS+ extends this by incorporating environment descriptions as structured prior knowledge. While these methods demonstrate the potential of language-guided optimization, they rely exclusively on scalar reward signals for feedback. BMPS builds upon this foundation by augmenting the optimization loop with multimodal behavioral diagnostics from vision-language models, enabling diagnosis-aware policy refinement beyond reward-only prompting.

For policy-gradient and value-based reinforcement learning methods, we include TRPO [27], PPO [28], A2C [20], and DQN [19].. For evolutionary strategies, we evaluate OpenAI-ES [26], which estimates gradients via population perturbations; CMA-ES [8], which adapts a full covariance matrix to model the search distribution; the classical (μ, λ)-ES, which selects the top μ individuals from λ offspring; and ARS [17], a simple random-search method that directly perturbs policy parameters. All baseline methods are trained for a fixed budget of 8×10^6 environment interaction steps to ensure fair comparison across approaches. Each run stops after 400 iterations, 8×10^6 environment steps including neighborhood rollouts, or when the prompt exceeds the LLM context window, whichever comes first.

LLM and VLM Setting. For all LLM-based methods, including BMPS, we use GPT-OSS-120B [23] as the LLM and Gemma 3 27B-IT [32] as the VLM. The temperature for both the LLM and VLM is set to 1.

5.2 Performance on Continuous Control Tasks

Table 1 reports results on seven continuous-control benchmarks. BMPS achieves the highest mean return on four environments (Mountain Car Continuous, Inverted Pendulum, Swimmer, Walker). On Reacher, its mean return is close to PPO's (−7.36 versus −7.32). However, on Hopper, BMPS reaches 703.97 ± 323.75, compared with CMA-ES at 1677.37 ± 612.72. On Inverted Double Pendulum, BMPS has a lower mean than TRPO (982.31 versus 3609.37) amid large run-to-run variability.

Variability also depends on the task and comparator. BMPS records 1000.00 ± 0.00 on Inverted Pendulum and a lower standard deviation than the evolutionary baselines on Swimmer (57.38, compared with 157.72 for CMA-ES). In contrast, its standard deviation on Inverted Double Pendulum is 2482.35, exceeding its mean of 982.31, and variability remains substantial on Hopper (323.75). These results do not establish a general reduction in variance or sensitivity to hyperparameters and initialization.

5.3 Performance on Discrete Control Tasks

Table 2 summarizes performance on six discrete-action benchmarks. BMPS achieves the highest mean return on Frozen Lake, Cliff Walking, Mountain Car, and Pong, and ties for best on Maze. On Navigation, BMPS has a higher mean than the other LLM-based and evolutionary methods but a lower mean than A2C, PPO, and TRPO. Its return is 2279.48 ± 1172.74, compared with TRPO's 4223.51 ± 19.70, indicating both a lower mean and substantial run-to-run variability.

Frozen Lake and Mountain Car illustrate gains on tasks with sparse or initially uninformative returns. On Frozen Lake, BMPS achieves 0.975 ± 0.025, compared with 0.15 for A2C and 0.16 for PPO. On Mountain Car, BMPS achieves -112.37 ± 2.58, compared with ProPS+ at -117.47 ± 23.35, with a higher mean and a lower standard deviation. These observations are consistent with behavioral feedback helping search on some tasks, but the return tables alone do not establish that it prevents local optima or premature convergence.

Table 1: Performance on continuous control environments. Policy-gradient baselines follow [38] (exact same setting).

Method	Mtn. Car Cont.	Inv. Pendulum	Inv. Dbl. Pendulum	Reacher	Swimmer	Hopper	Walker
Policy Gradient Methods							
A2C	74.10 ± 7.44	155.15 ± 53.79	102.90 ± 32.04	-53.48 ± 21.42	39.40 ± 6.54	123.49 ± 86.46	355.84 ± 154.02
PPO	78.16 ± 5.32	218.65 ± 129.31	108.60 ± 4.12	**-7.32 ± 0.38**	37.31 ± 7.19	351.75 ± 157.71	469.78 ± 159.17
TRPO	1.18 ± 3.54	571.31 ± 358.88	**3609.37 ± 4000.04**	-8.93 ± 1.39	52.96 ± 18.86	716.90 ± 385.20	519.38 ± 73.15
LLM-based Methods							
ProPS	99.09 ± 0.18	819.17 ± 362.37	125.89 ± 32.67	-7.95 ± 0.58	151.64 ± 56.03	407.86 ± 301.35	283.88 ± 65.17
ProPS+	97.92 ± 1.90	882.19 ± 238.13	144.52 ± 69.90	-7.61 ± 0.35	210.65 ± 58.89	421.35 ± 294.93	318.46 ± 45.07
Evolutionary Computing Methods							
OpenAI-ES	9.80 ± 29.72	364.94 ± 417.36	91.78 ± 4.54	-244.79 ± 335.38	114.22 ± 95.34	975.21 ± 174.19	-6.52 ± 4.67
CMA-ES	30.90 ± 44.70	604.07 ± 484.97	1052.25 ± 2769.83	-9.60 ± 1.16	205.14 ± 157.72	**1677.37 ± 612.72**	145.67 ± 297.01
(μ, λ)**-ES**	37.95 ± 47.08	477.49 ± 441.24	113.10 ± 99.16	-21.71 ± 31.56	167.73 ± 126.58	1224.61 ± 404.43	-13.30 ± 8.09
ARS	9.59 ± 29.81	216.80 ± 393.03	151.46 ± 229.83	-39.12 ± 49.89	51.70 ± 88.47	283.13 ± 339.84	-41.94 ± 22.69
Ours							
BMPS	**99.13 ± 0.10**	**1000.00 ± 0.00**	982.31 ± 2482.35	-7.36 ± 0.66	**289.47 ± 57.38**	703.97 ± 323.75	**552.02 ± 175.39**

Table 2: Performance on discrete control environments. Policy-gradient baselines follow [38] (exact same setting).

Method	Frozen Lake	Cliff Walking	Maze	Mtn. Car	Navigation	Pong
Policy Gradient Methods						
A2C	0.15 ± 0.03	-172.30 ± 14.49	**0.97 ± 0.00**	-200.00 ± 0.00	3670.87 ± 124.29	0.57 ± 0.14
DQN	0.13 ± 0.03	-320.93 ± 29.60	0.86 ± 0.22	-194.36 ± 1.47	-127.59 ± 7.13	0.56 ± 0.02
PPO	0.16 ± 0.02	-94.35 ± 3.96	0.96 ± 0.00	-200.00 ± 0.00	4127.43 ± 24.29	2.29 ± 0.91
TRPO	0.22 ± 0.05	-66.60 ± 13.61	**0.97 ± 0.00**	-200.00 ± 0.00	**4223.51 ± 19.70**	1.36 ± 1.05
LLM-based Methods						
ProPS	0.29 ± 0.09	-100.00 ± 0.00	**0.97 ± 0.00**	-190.58 ± 28.28	2003.38 ± 1094.46	1.89 ± 0.98
ProPS+	0.925 ± 0.026	-69.73 ± 8.51	**0.97 ± 0.00**	-117.47 ± 23.35	1863.72 ± 1240.19	2.65 ± 0.43
Evolutionary Computing Methods						
OpenAI-ES	0.22 ± 0.15	-540.10 ± 922.00	-1.11 ± 0.00	-200.00 ± 0.00	136.75 ± 768.11	1.11 ± 0.62
CMA-ES	0.16 ± 0.19	-235.30 ± 201.40	-1.11 ± 0.00	-200.00 ± 0.00	575.25 ± 1273.55	1.07 ± 0.97
(μ, λ)**-ES**	0.24 ± 0.07	-108.30 ± 11.20	-1.11 ± 0.00	-200.00 ± 0.00	669.63 ± 1216.00	0.77 ± 0.86
ARS	0.16 ± 0.11	-271.80 ± 161.50	-0.90 ± 0.62	-200.00 ± 0.00	-113.33 ± 94.17	0.22 ± 0.07
Ours						
BMPS	**0.975 ± 0.025**	**-64.35 ± 5.98**	**0.97 ± 0.00**	**-112.37 ± 2.58**	2279.48 ± 1172.74	**2.96 ± 0.11**

5.4 Ablation Studies

To better understand how different components of BMPS contribute to search performance, we conduct ablation studies on *Swimmer* and discrete *MountainCar*, which serve as representative benchmarks for continuous and discrete control, respectively.

Effect of Environment Prior Knowledge. We first examine the contribution of textual prior knowledge about the environment by comparing the full method with a variant whose LLM prompt omits the environment description. In Table 3, full BMPS has a higher mean return and lower standard deviation on both tested tasks.

Table 3: Ablation of environment prior knowledge.

Method	*Swimmer*	*MountainCar*
w/o environment prior	206.325 ± 103.107	-142.055 ± 40.751
BMPS	**309.36 ± 51.83**	**-112.01 ± 2.85**

These results suggest a benefit from environment context in these settings.

Table 4: Ablation of contextual and anchor-policy components.

Method	*Swimmer*	*MountainCar*
w/o optimization history	270.95 ± 21.63	-133.95 ± 28.20
w/o neighbor sampling	202.07 ± 59.38	-135.26 ± 33.75
w/o current anchor	253.14 ± 70.36	-125.75 ± 25.72
w/o best anchor	258.55 ± 84.58	-141.09 ± 34.42
w/o worst anchor	268.02 ± 90.37	-117.7 ± 20.69
BMPS	**309.36 ± 51.83**	**-112.01 ± 2.85**

Table 5: Ablation study on the number of neighbors in BMPS.

Method	*Swimmer*	*MountainCar*
0 neighbor	202.07 ± 59.38	-135.26 ± 33.75
3 neighbors	215.35 ± 90.08	-140.93 ± 36.88
7 neighbors	275.64 ± 87.53	-119.71 ± 20.23
BMPS (5 neighbors)	**309.36 ± 51.83**	**-112.01 ± 2.85**

Effect of Visual Feedback and Selection of Anchor Policies. We next study the contribution of the multimodal diagnosis module by removing individual components from the full method. As shown in Table 4, we evaluate variants without optimization history, without neighbor sampling, and without each anchor policy type (current anchor, best anchor, and worst anchor), in order to identify which sources of contextual and behavioral information are most important for stable policy improvement. Here, optimization history refers to the summary of previous policy proposals and their observed returns provided to the LLM prompt. Removing this component tests whether iterative search in BMPS benefits from remembering past trials rather than relying only on the current round of multimodal feedback.

Effect of the Number of Neighbors. Finally, we vary the number of neighboring policies sampled around each anchor. Among the tested settings in Table 5, five neighbors yields the highest mean return and lowest standard deviation on both environments, and we use this setting in the main experiments. Performance does not improve monotonically with the number of neighbors: three neighbors has a lower mean than zero, while seven has a lower mean than five. These comparisons support the choice of five neighbors for the two tested tasks, but do not establish a general optimum or identify whether redundant or conflicting feedback explains the differences.

5.5 Case Study: How Behavioral Feedback Guides the Search

Figure 2 traces how the behavioral landscape enters the proposals of one BMPS run on *Swimmer*. The initial anchor folds into an L-shape and holds it while drifting ($R = 26.8$), and the VLM describes exactly this posture. At iteration 3, the landscape contrasts low-return rollouts that "fold into an L-shape" and drift with a neighbor that keeps "bending and straightening its body" ($R = 108.6$), and the LLM moves toward that neighbor while citing vertical drift and rotor control in its rationale. The resulting policy θ_3 folds and unfolds with large amplitude and reaches $\hat{J} = 116.6$. At iteration 13, the descriptions report oscillation with over-corrections, and the LLM adjusts the bias terms to "stabilize action magnitude, reducing chaotic over-corrections while preserving strong forward thrust"; θ_{13} settles into a repetitive bending pattern and reaches $\hat{J} = 243.5$, close to the best return of the run. The rationales also rely on the returns of the neighbors, so the trace illustrates how behavioral and numerical evidence are combined in a proposal rather than isolating the effect of either.

6 Conclusion

We introduced BMPS, a multimodal policy search framework that enriches LLM-based optimization with structured visual feedback from a VLM. By combining reward histories with trajectory-level behavioral diagnostics, BMPS provides behavioral evidence for policy proposals. In the 13-task evaluation, it achieves the best average rank and higher mean returns than ProPS and ProPS+ on 12 tasks, tying on Maze. Its advantages over policy-gradient and evolutionary methods depend on the task: it trails leading baselines on Hopper, Navigation, and Inverted Double Pendulum, and substantial run-to-run variability remains on some tasks.

A key limitation is token efficiency: augmenting prompts with optimization history, neighbor comparisons, and VLM feedback is substantially more costly than reward-only baselines, and scaling to longer trajectories or richer observations may require aggressive prompt compression. Looking ahead, an important direction is formalizing the optimization loop as a sequential decision-making problem, which would provide a principled foundation for managing the exploration–exploitation trade-off and reducing context overhead over time.

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